

FUNCTION SPACES

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Chapter 1

FUNCTION SPACES

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Quizzes 1 and 2 will account for 10% and 15% of the final grade respectively. The mid-term will account for 25% of the final grade, and the final exam will account for 50%.

1.1 An Introduction

Suppose $\mathbb{K}$ is a field, and X is a topological space. Then the set of all $\mathbb{K}$ -valued functions on X , denoted by $\mathbb{K}^X$, is a vector space over $\mathbb{K}$. If we let $X = \{1, 2, \dots, n\}$, then $\mathbb{K}^X \cong \mathbb{K}^n$, the n -dimensional vector space over $\mathbb{K}$. In this course, we will be primarily concerned with the case when $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Consider the map

$$D : \mathbb{K}^{\{1, 2, \dots, n\}} \rightarrow \mathbb{K}^{\{1, 2, \dots, n-1\}}, \quad (Df)(k) = f(k+1) - f(k) \text{ for } 1 \leq k \leq n-1, \quad (Df)(n) = f(1) - f(n). \quad (1.1)$$

Here, D acts as a discrete analogue of the derivative operator by taking finite differences. We can split this finite difference map as the difference of the left shift operator and the identity operator as $D = S - I$. Since S is the left shift operator, we have $S^n = I$ so the eigenvalues of S are a subset of the n -th roots of unity. Let λ be such an eigenvalue. Then a corresponding eigenvector can be given as

$\begin{pmatrix} 1 \\ \lambda \\ \vdots \\ \lambda^{n-1} \end{pmatrix}$. If $\lambda = e^{2\pi i/n}$, then we can denote the corresponding eigenvector of λ^k as χ_k . In fact, in the

basis $\{\chi_1, \dots, \chi_{n-1}, \chi_n\}$, D is a diagonal matrix. Such a basis is termed a *discrete Fourier basis*, and the coordinate vector of D with respect to this basis is called the *discrete Fourier transform* of D . The Fourier transform diagonalizes the finite differential operator. We can view χ_k as a function $\chi_k : \{1, 2, \dots, n\} \rightarrow \mathbb{C}$ defined by $\chi_k(j) = e^{2\pi i k(j-1)/n}$ for $1 \leq j \leq n$.

Recall the Hermitian inner product on $\mathbb{C}^n$ given by $\langle x, y \rangle = \sum_{i=1}^n x_i \overline{y_i}$. Viewing $\mathbb{C}^n$ as the function space $\mathbb{C}^{\{1, 2, \dots, n\}}$, we can endow it with the inner product $\langle f, g \rangle = \sum_{k=1}^n f(k) \overline{g(k)}$, which gives the 2-norm $\|f\|_2 = \sqrt{\sum_{i=1}^n |f(i)|^2}$. The discrete Fourier basis is an orthogonal basis with respect to this inner product. Normalizing via division by $\sqrt{n}$ makes it an orthonormal basis. Thus, if we have $f \in \mathbb{C}^{\{1, 2, \dots, n\}}$ which we wish to write as $f = \sum_{j=1}^n a_j \chi_j$, then we can compute the coefficients a_j as

$$a_j = \frac{1}{n} \langle f, \chi_j \rangle = \frac{1}{n} \sum_{r=1}^n f(r) e^{-2\pi i j(r-1)/n}. \quad (1.2)$$

This is the discrete Fourier transform, as before. If we look at the norm, then

$$\|f\|_2^2 = \left\| \sum_{j=1}^n a_j \chi_j \right\|_2^2 = n \sum_{j=1}^n |a_j|^2 \implies \frac{1}{n} \|f\|_2^2 = \sum_{j=1}^n |a_j|^2. \quad (1.3)$$

This is known as *Plancherel's theorem*.

All of the above can be generalized to the continuous case; let $L^2[0, 1]$ be the space of square integrable functions on $[0, 1]$. If $(a_n)_{n \in \mathbb{Z}}$ is the Fourier series of f , then $\int_0^1 |f(x)|^2 dx = \sum_{n \in \mathbb{Z}} |a_n|^2$, which is the continuous analogue of Plancherel's theorem.

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